Gao et al. BMC Pediatrics (2026) 26:112
https://doi.org/10.1186/s12887-025-06465-z

BMC Pediatrics

RESEARCH Open Access

The role of executive function for differentiating symptoms of ADHD in preschoolers

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Abstract

Background Attention-deficit/hyperactivity disorder (ADHD) is a chronic neurodevelopmental disorder associated with broad functional impairments in children. As a critical cognitive ability, executive function (EF) undergoes rapid development during preschool years. This study investigates the discriminative role of EF in identifying ADHD symptoms amid challenges in early assessment.

Methods A cohort of 141 preschoolers aged 4–5 years with ADHD was assessed with a battery of standardized instruments: the Behavior Rating Inventory of Executive Function-Preschool Edition (BRIEF-P) for EF, the Developmental Coordination Disorder Questionnaire-Chinese Version (DCDQ-C) for motor coordination, the Swanson, Nolan and Pelham, Version-IV Scale (SNAP-IV) for ADHD symptoms, the Parents' Evaluation on Preschooler's Social Skills (PEOPSS) and the Early Childhood Communication Interest Questionnaire.

Results Analyses using correlation tests, multiple linear regression, and receiver operating characteristic (ROC) curves revealed that ADHD symptoms were significantly associated with EF, motor coordination, social skills, and communication interests. ROC analyses identified differential links between specific EF components and symptom domains: Working memory strongly discriminated attention deficit (AD) (AUC = 0.854), as did the Inhibitory Self-Control Index (ISCI) for oppositional defiant symptoms (AUC = 0.852), while Plan/Organize demonstrated a more moderate capacity to discriminate hyperactivity/impulsivity (AUC = 0.719). Regression models confirmed EF as a robust predictor of ADHD symptoms.

Conclusion These results indicate that EF metrics can effectively distinguish ADHD symptoms in Chinese preschoolers.

Keywords ADHD, Executive function, Preschooler

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Introduction

Attention-deficit/hyperactivity disorder (ADHD) is a common chronic neurodevelopmental disorder that persists throughout life. Its core symptoms include inattention, hyperactivity, and impulsivity that are developmentally inappropriate. The global prevalence of ADHD in children is about 7.2% [1, 2]. According to 2016 national survey data from the United States, 2.4% of preschool children aged 2 to 5 had an ADHD diagnosis [3]. While its prevalence in China is estimated at 6.26%, affecting approximately 23 million children and adolescents [4]. This disorder adversely impacts multiple developmental domains, including academic achievement, behavioral functioning, peer and family relationships, and cognitive-emotional development [5, 6].

The preschool period marks a phase of rapid development in attention, behavioral control, and cognitive function [7]. Studies have found that preschoolers with ADHD have deficits in multiple functional domains, especially in activity level, self-regulation and executive function (EF) [8–11], that often persist into adulthood [4]. Research indicates that the anterior portion of the brain, particularly the prefrontal cortex (PFC), serves as the core neural substrate for executive processes [12]. Studies have demonstrated that individuals with ADHD exhibit reduced activation in the PFC during EF tasks, along with potential functional and structural abnormalities in this region [13]. The protracted and organized maturation of the PFC underlies the development of EF, and accounts for its progression from simple and unstable to increasingly complex and stable during childhood. The preschool period represents a critically dynamic phase for prefrontal development, marking the emergent stage of EF. Therefore, investigating ADHD during this preschool stage contributes significantly to clarifying the characteristic profiles of its symptoms and functional impairments.

EF refers to a series of advanced cognitive functions that interact synergistically in the process of goal setting, planning, and implementation by the individual, and which essentially regulate other cognitive processes to generate coordinated, orderly, and goal-directed behaviors [14, 15]. Theoretical models over the past several decades propose that EF is comprised of core components that emerge early in development and continually build over time, forming the complex, higher-order cognitive abilities essential for daily functioning [16]. Therefore, targeting EF in preschoolers with ADHD is important for their lifelong development, as well as early recognition of symptoms [17, 18].

The assessment of EF deficits in children often relies on parent-reported behavioral rating scales. The Behavior Rating Inventory of Executive Function (BRIEF) is a widely recognized tool for this purpose in children with ADHD [19]. The preschool version (BRIEF-P), developed by Gioia et al., is designed to assess EF through observable behavioral performance of specific cognitive processes, which had a good internal consistency (0.80 to 0.97) and temporal stability (0.65 to 0.94) in the preschool population [20, 21]. Integrating such measures enables a more comprehensive evaluation of ADHD and facilitates targeted interventions to improve developmental outcomes [22].

Previous studies have explored the role of EF in assessing ADHD symptoms in preschoolers [23–25]. One study finds that deficits in EF, as measured by both parent/teacher reports and performance on an EF battery, are significantly associated with inattention. In contrast, parent/teacher reports of EF, but not performance-based measures, are associated with hyperactivity [23]. Furthermore, inattentive symptoms consistently demonstrate a stronger link to EF deficits than hyperactive/impulsive symptoms [24]. However, some studies propose that EF may contributes less prominently to ADHD behaviors than commonly assumed in early childhood [26]. For younger children, caregiver reported temperament features may have greater clinical utility in the early detection of ADHD symptoms compared to EF [27]. Given that EF represents a core neuropsychological mechanism in ADHD, clarifying these discrepant findings in the preschool period is critical.

Although ADHD is associated with diverse neurocognitive impairments, the extent to which EF deficits underlie the progression of ADHD symptoms remains unclear. Furthermore, few studies have systematically examined the discriminative accuracy of specific EF indices for symptom domains in preschoolers. This study aims to investigate the role of EF in differentiating ADHD symptoms in order to better characterize the clinical manifestations and functional impairments in the early developmental stages of children with ADHD.

Methods

Participants

The children were recruited from the outpatient clinic of the Department of Psychology at the Children's Hospital of Zhejiang University School of Medicine between January 2021 and December 2021. The inclusion criteria were: (1) aged 4–5 years; (2) a clinical diagnosis of ADHD by psychiatrists or psychologists according to Diagnostic and Statistical Manual of Mental Disorders 5th edition (DSM-5); (3) parents/guardians had at least primary school education. Children were excluded if they had: (1) severe neurological or developmental disorders such as cerebral palsy, pervasive developmental disorder, or epilepsy; (2) an intellectual quotient (IQ) < 75; (3) comorbid psychiatric disorders such as schizophrenia, autism

spectrum disorder, or tic disorders; (4) genetic syndromes or other severe neurobehavioral disorders.

This study was conducted in accordance with the Declaration of Helsinki and approved by the Institutional Research Ethics Board of Children's Hospital, Zhejiang University School of Medicine (Approval No. 2023-IRB-0318-P-01). The requirement for informed consent was waived due to the exclusive use of anonymized historical data with minimal risk, as defined under Chinese biomedical research ethics regulations.

Procedure

All participants underwent professional interviews and evaluations, followed by screening based on the predefined inclusion and exclusion criteria. Participants and caregivers completed a battery of assessments: BRIEF-P, Developmental Coordination Disorder Questionnaire-Chinese Version (DCDQ-C), Swanson, Nolan and Pelham, Version-IV Scale (SNAP-IV), Parents' Evaluation on Preschooler's Social Skills (PEOPSS) and Early Childhood Communication Interest Questionnaire. After excluding invalid BRIEF-P protocols, 141 cases were retained, yielding a valid response rate of 98.6%.

Measures

Parent-report measures (e.g., BRIEF-P) were selected over performance-based tasks for this study of preschool-aged children for several key reasons. First, they capture behaviors across diverse real-world settings (e.g., home, community), offering ecological validity that single-session lab tasks may lack. Second, young children's limited attention spans, comprehension, and motivation can make performance on structured EF tasks highly variable and susceptible to non-compliance, potentially obscuring their true abilities. Lastly, parent reports efficiently summarize extensive observational experiences, providing a practical and comprehensive assessment of everyday executive functioning.

(1) BRIEF-P: This 63-item scale assesses everyday behaviors in children aged 2 years 0 months to 5 years 11 months [28]. It measures five subdomains: Inhibit, Shift, Emotional control, Working memory, and Plan/Organize. These are combined into three composite indices: Inhibitory Self-Control Index (ISCI, Inhibit and Emotional control), Flexibility Index (FI, Shift and Emotional control) and Emergent Metacognition Index (EMI, Working memory and Plan/Organize). And overall Global Executive Composite (GEC) score is also derived. The BRIEF-P has demonstrated validated effectiveness in assessing EF among Chinese preschool-aged children [29]. In this study, the BRIEF-P demonstrated high internal consistency (Cronbach's $\alpha = 0.933$).

(2) DCDQ-C: This questionnaire is completed by parents or teachers to assess the child's motor coordination skills in daily life and learning [30]. The 15-item questionnaire comprises three subscales: Motor Control, Fine Motor, and Motor Coordination. A total score below 46 was used as the cutoff for suspecting developmental coordination disorder (DCD). In this study, the scale's Cronbach's α was 0.893.

(3) SNAP-IV: The 26-item scale consists of three subscales: attention deficit, hyperactivity/impulsivity, and oppositional defiant symptoms, and is completed by parents [31]. In this study, the scale's Cronbach's α was 0.886.

(4) PEOPSS: The 32-item scale consists of two domains: Self-control (comprising Self-management, Persistence, Consciousness, and Delayed gratification) and Initiative (comprising Spontaneity, Expressiveness, Pioneership, and Helpfulness). Higher scores indicate more developed social skills. The full questionnaire is detailed in Supplementary Materials (Table S1). In this study, the PEOPSS demonstrated high internal consistency (Cronbach's $\alpha = 0.897$).

(5) Early Childhood Communication Interest Questionnaire: It consists of 9 items, with higher scores indicating greater interest and better ability to communicate (see Table S2 for full items). In this study, the scale's Cronbach's α was 0.505.

Statistical analysis

We used SPSS 23.0 (IBM, Armonk, NY, USA) for statistical analysis. The data were presented as mean ± standard error (M ± SEM). We conducted a multiple linear regression to identify predictors of ADHD symptoms using the SNAP-IV. Correlational analyses were conducted to assess the relationship between the DCDQ-C, BRIEF-P, PEOPSS, Early Childhood Communication Interest Questionnaire and the SNAP-IV. The factors that may affect AD, hyperactivity/impulsivity and oppositional defiant symptoms in preschool children were analyzed by binary logistic regression, and the statistically significant factors were included in ROC analysis. For each ROC analysis, we calculated the area under the curve (AUC), sensitivity and specificity. AUC values > 0.80 were classified as excellent; 0.70–0.80 as good; 0.60–0.70 as acceptable, and < 0.60 as poor with limited clinical utility. Statistical significance for AUC was set at $p < .05$.

Results

Correlation analysis

Table 1 summarized the demographic characteristics of the study sample. A total of 141 preschool children (4.90 ± 0.52 years) with ADHD were included, comprising 119 males (84.4%) and 22 females (15.6%).

Correlation analyses were performed to examine any potential associations among demographic variables (Fig. 1). For SNAP-IV-AD, scores showed positive correlations with BRIEF-P-Working memory (r = .59, p < .001) and BRIEF-P-ISCI (r = .37, p < .001). For SNAP-IV-hyperactivity/impulsivity, scores were positively correlated with the BRIEF-P-GEC (r = .55, p < .001), BRIEF-P-FI (r = .36, p < .001), BRIEF-P-Plan/Organize (r = .51, p < .01), and Communication Interests (r = .25, p < .05); and were negatively correlated with PEOPSS-Self-management (r = −.35, p < .001). For SNAP-IV-oppositional defiant symptoms, scores demonstrated positive correlations with BRIEF-P-ISCI (r = .61, p < .001) and BRIEF-P-Inhibit (r = .50, p < .001), and a negative correlation with PEOPSS-Self-management (r = −.39, p < .001).

Table 1 Descriptive statistics

Screening measures	Mean ± SEM
Sex (male: female)	119:22
Age	4.90 ± 0.52
Parental education level (years)	14.12 ± 2.70
ADHD symptoms (SNAP-IV)	
AD	1.39 ± 0.41
Hyperactivity/impulsivity	1.58 ± 0.51
Oppositional defiant symptoms	1.11 ± 0.44
Total	1.37 ± 0.37
Developmental coordination (DCDQ-C)	
Motor control	19.43 ± 4.91
Fine motor	12.69 ± 4.17
Motor coordination	15.45 ± 4.35
Total	47.57 ± 11.55
Communication interests in early childhood	
Communication interests	8.01 ± 1.21
Executive functions (BRIEF-P)	
Inhibit	0.98 ± 0.33
Shift	0.53 ± 0.29
Emotional control	0.65 ± 0.37
Working memory	0.90 ± 0.30
Plan/Organize	0.82 ± 0.32
ISCI	0.86 ± 0.30
FI	0.59 ± 0.29
EMI	0.87 ± 0.28
GEC	0.81 ± 0.26
Social skills (PEOPSS)	
Self-management	9.34 ± 2.25
Persistence	16.38 ± 3.57
Consciousness	9.22 ± 2.29
Delayed gratification	13.85 ± 3.05
Self-control	48.79 ± 7.49
Spontaneity	10.70 ± 2.31
Expressiveness	13.54 ± 2.91
Pioneership	16.79 ± 3.52
Helpfulness	13.97 ± 3.49
Initiative	54.99 ± 10.63

SNAP-IV Swanson Nolan and Pelhanm, Version- IV Scale, *AD* Attention deficit, *DCDQ-C* Developmental Coordination Disorder Questionnaire-Chinese version, *BRIEF-P* Behavior Rating Inventory of Executive Function -Preschool version, *ISCI* Inhibitory Self-Control Index, *FI* Flexibility Index, *EMI* Emergent Metacognition Index, *GEC* Global Executive Composite, *PEOPSS* Parents' Evaluation on Preschooler's Social Skills

Multiple linear regression analysis of SNAP-IV

We conducted a stepwise linear regression analysis on the AD, hyperactivity/impulsivity, and oppositional defiant symptoms scores of SNAP-IV using clinical scale indicators (Table 2). BRIEF-P-Working memory (β = 0.62, p < .001), PEOPSS-Pioneership (β = 0.35, p < .001), and PEOPSS-Expressiveness (β = −0.20, p < .05) significantly predicted AD score (F(3,137) = 32.24, p < .001); BRIEF-P-GEC(β = 0.78, p < .001), PEOPSS-Self-management(β = −0.21, p < .05), PEOPSS-Pioneership(β = 0.19, p < .05), BRIEF-P-FI(β = −0.33, p < .05), Age(β = 0.15, p < .05), and Communication interests(β = 0.14, p < .05) significantly predicted hyperactivity/impulsivity score (F(6,134) = 17.20, p < .001); BRIEF-P-ISCI(β = 0.99, p < .001), PEOPSS-Self-management(β = −0.22, p < .05), BRIEF-P-Inhibit(β = −0.46, p < .05), and Communication interests(β = −0.14, p < .05) significantly predicted oppositional defiant symptoms score (F(4,136) = 29.50, p < .05).

Identification AD, hyperactivity/impulsivity and oppositional defiant symptoms of SNAP-IV in ADHD in preschoolers by EF

We used stepwise logistic regression to construct the models and then assessed the ability of the EF indicators to identify group membership of SNAP-IV (AD/hyperactivity/impulsivity/oppositional defiant symptoms, i.e., average score > 1 or not). For identification of SNAP-IV-AD, BRIEF-P-Working Memory yielded an AUC of 0.854 (85.1% sensitivity, 70.4% specificity), while BRIEF-P-ISCI yielded an AUC of 0.591 (42.1% sensitivity, 74.1% specificity). For identification of SNAP-IV-hyperactivity/impulsivity, BRIEF-P-Plan/Organize achieved an AUC of 0.719 (63.9% sensitivity, 68.2% specificity). For identification of SNAP-IV-oppositional defiant symptoms, BRIEF-P-ISCI showed an AUC of 0.852 (66.2% sensitivity, 86.3% specificity). Complete results are presented in Table 3; Fig. 2.

Discussion

This study aimed to provide a comprehensive characterization of preschoolers with ADHD by examining their symptomatic profiles, EF deficits, and the interplay

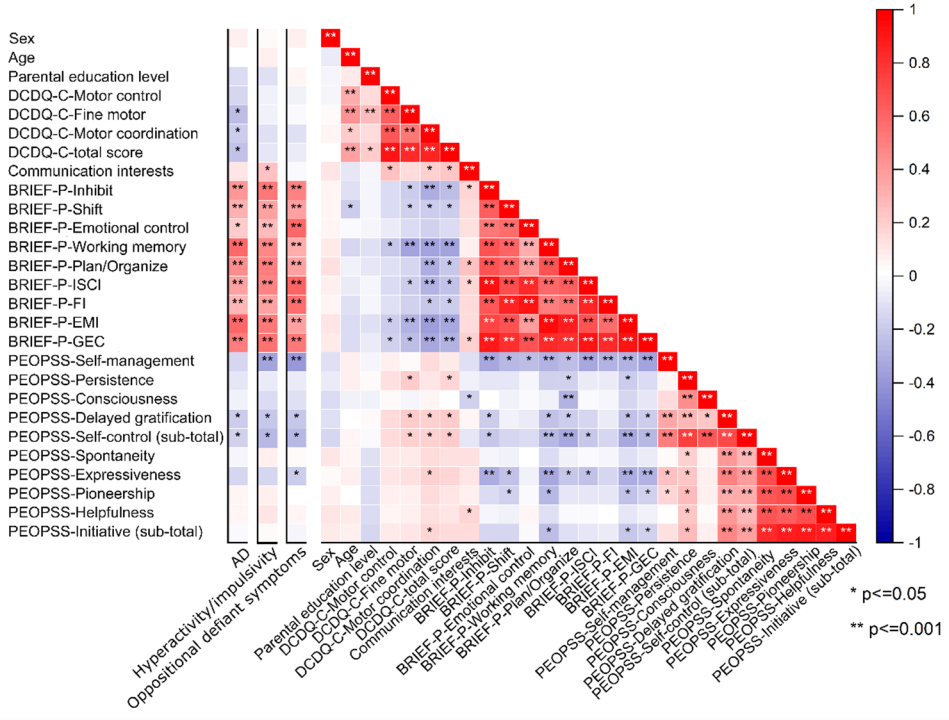


Fig. 1 Correlation analysis of each factor. Correlation analyses were performed to examine potential associations among the variables (Fig. 1). SNAP-IV-AD subscales were negatively correlated with the DCDQ-C-Fine motor ($p < .05$) and Motor coordination ($p < .05$). Furthermore, SNAP-IV scores were negatively correlated with some PEPOSS indices, while they were positively correlated with all BRIEF-P indices. Regarding EF, multiple indices from the BRIEF-P were correlated with the DCDQ-C, PEOPSS, and communication interests

Table 2 Multiple linear regression analysis of SNAP-IV scores in preschoolers with attention-deficit/hyperactivity disorder

	Variable	B	Standard error	β	t	*P*
AD	Constant	0.31	0.19		1.61	0.109
	BRIEF-P-Working memory	0.85	0.10	0.62	8.93	<0.001
	PEOPSS-Pioneership	0.04	0.01	0.35	3.86	<0.001
	PEOPSS-Expressiveness	−0.03	0.01	−0.20	−2.13	0.035
hyperactivity/impulsivity	Constant	−0.53	0.47		−1.13	0.261
	BRIEF-P-GEC	1.51	0.24	0.78	6.41	<0.001
	PEOPSS-Self-management	−0.05	0.02	−0.21	−2.97	0.004
	PEOPSS-Pioneership	0.03	0.01	0.19	2.89	0.005
	BRIEF-P-FI	−0.58	0.20	−0.33	−2.83	0.005
	Age	0.15	0.06	0.15	2.30	0.023
	Communication interests	0.06	0.03	0.14	2.11	0.037
oppositional defiant symptoms	Constant	1.29	0.25		5.17	<0.001
	BRIEF-P-ISCI	1.42	0.23	0.99	6.10	<0.001
	PEOPSS-Self-management	−0.04	0.01	−0.22	−3.32	0.001
	BRIEF-P-Inhibit	−0.60	0.21	−0.46	−2.85	0.005
	Communication interests	−0.05	0.02	−0.14	−2.17	0.032

SNAP-IV Swanson Nolan and Pelhanm, Version- IV Scale; *AD* Attention deficit, *BRIEF-P* Behavior Rating Inventory of Executive Function -Preschool version, *GEC* Global Executive Composite, *FI* Flexibility Index, *ISCI* Inhibitory Self-Control Index, *PEOPSS* Parents' Evaluation on Preschooler's Social Skills

between them at an early developmental stage. Findings from this study showed that the BRIEF-P Working memory and ISCI subscales were associated with the attention deficit subscale of SNAP-IV, the BRIEF-P Plan/Organize subscale was significantly associated with the hyperactivity/impulsivity subscale of SNAP-IV, and the BRIEF-P ISCI subscale was significantly associated with the oppositional defiant symptom subscale of SNAP-IV. Furthermore, poorer developmental coordination was strongly associated with inattention, while deficits in social skills and communication interests were also linked to ADHD symptomatology.

Table 3 Results of differentiating symptoms of ADHD in preschoolers by EF

Identification group	Identification criteria	Sensitivity (%)	Specificity (%)	AUC [95%CI]
AD	BRIEF-P-Working memory	85.1	70.4	0.854[0.783,0.925]
	BRIEF-P-ISCI	42.1	74.1	0.591[0.474,0.709]
hyperactivity/impulsivity	BRIEF-P-Plan/Organize	63.9	68.2	0.719[0.603,0.835]
oppositional defiant symptoms	BRIEF-ISCI	66.2	86.3	0.852[0.791,0.913]

AD Attention deficit, *BRIEF-P* Behavior Rating Inventory of Executive Function -Preschool version, *ISCI* Inhibitory Self-Control Index, *AUC* Area under the curve, *CI* Confidence interval

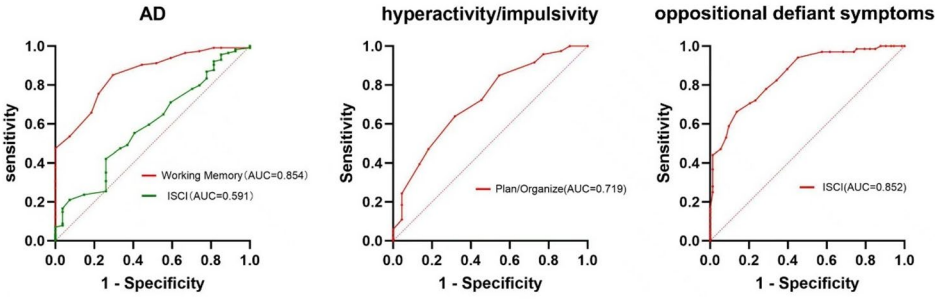


Fig. 2 ROC analysis of ADHD symptoms

Working memory and ISCI in the identification of attention deficit

We found that the BRIEF-P Working memory and ISCI (comprising Inhibit and Emotional control) subscales differentiated preschoolers on the SNAP-IV attention deficit subscale, with Working memory demonstrating superior discriminative validity (Table 3; Fig. 2). Although the behavioral expression of ADHD may be influenced by cultural norms, growing neurocognitive evidence indicates that its core cognitive deficits - particularly working memory dysfunction - demonstrate remarkable consistency across diverse cultural contexts [16]. Working memory, a system for the temporary storage and processing of information during cognitive tasks, guides behavior via the mid-lateral prefrontal cortex. Studies have revealed that 89% of children with ADHD exhibit impairment in at least one executive function, and those with working memory deficits show more severe inattention and hyperactivity/impulsivity symptoms [32]. Longitudinal evidence indicates that superior working memory predicts better long-term ADHD outcomes. Furthermore, research specifically links working memory deficits to inattentive symptoms and subsequent functional impairment, but not to hyperactive/impulsive symptoms [33, 34]. One study with up to 13 years of follow-up found that preschool working memory and reaction time variability predicted future inattention and were linked to later academic performance [17]. Thus, impairment in working memory is likely to be directly manifested as significant inattentive symptoms, regardless of cultural background.

Inhibitory control – defined as the capacity to suppress prepotent responses and filter irrelevant stimuli – constitutes a core component of EF, that regulates context-appropriate behavioral. Compared to typically developing children, those with ADHD exhibit significant deficits in response inhibition, working memory, and response variability. Crucially, longitudinal data suggest that while most EF differences diminish by adolescence, response inhibition deficits persist into puberty, highlighting their enduring neurodevelopmental impact [35]. Furthermore, one study demonstrated that simple inhibition and selective attention at age 5 independently predict complex inhibition and working memory at age 6. The results also showed that, even in young children, EF components primarily predicted symptoms of inattention rather than hyperactivity/impulsivity [36]. Thus, inhibitory control appears to be a developmental precursor to later development and more complex EF deficits in children with ADHD and is primarily associated with inattention.

Prior research indicats that approximately half of children with ADHD experience emotion regulation difficulties, which in turn lead to greater distress and impairment than ADHD symptoms alone [37, 38]. A significant and robust association exists between ADHD and emotional reactivity, meaning that adolescents with ADHD are more likely to experience intense emotions (both negative and positive) and exhibit stronger reactions to stressful and/or frustrating situations [38]. Bunford et al. reported that emotional dysregulation in adolescents is associated with ADHD and does not differ by subtype [39]. Furthermore, three measures of emotional functioning—anger regulation, anger recognition, and the regulation of happiness/exuberance—significantly discriminate children with ADHD from typically developing

peers, with emotion regulation showing independent effects on ADHD identification [40]. Notably, one study found that the regulation of happiness/exuberance was significantly related to inattention, suggesting that emotion regulation deficits constitute an important aspect of ADHD [17]. Another study supports this, showing that preschool ADHD symptoms are associated not only with pronounced and often occurring negative emotions, but also specifically with the regulation of negative and positive emotions. Importantly, the significant independent effects of emotion regulation on ADHD symptoms outweigh those of cognitive and motivation-based regulation [41]. However, research on the synergistic effects of inhibition and emotion regulation on inattentive symptoms remains lacking.

Plan/organize in the identification of hyperactivity/impulsivity

Our study demonstrated that the BRIEF-P Plan/Organize subscale differentiate the hyperactivity/impulsivity subscale of the SNAP-IV (Table 3; Fig. 2). This finding extends the current literature by establishing the discriminative validity of planning and organization skills for early ADHD identification. Plan/Organize involves the ability to anticipate future events, set goals, and plan out steps to accomplish tasks. However, these skills are challenging to assess in preschoolers due to their limited cognitive maturity. Behavioral observation protocols (e.g., the Tower of London) may have higher sensitivity in capturing planning and organization skills compared with parent-report questionnaires, though further refinement of scoring criteria and ecological validity in naturalistic settings remains necessary.

ISCI in the identification of oppositional defiant symptoms

As shown in Tables 2 and 3; Fig. 2, the BRIEF-P ISCI subscale is strongly associated with the oppositional defiant symptoms subscale of SNAP-IV in preschool children. EF, particularly inhibitory control, is associated with externalizing behaviors from the preschool period. Meta-analyses confirm this link and reveal differential associations across EF domains: working memory and cognitive flexibility show small effect sizes, whereas inhibition demonstrates a medium effect size [42]. Consistent with prior evidence, our results indicate that impaired inhibitory control predicts elevated problematic behaviors. Moreover, previous studies suggest that inhibitory deficits underlie the development of externalizing problem behaviors and exacerbate core ADHD symptoms [43]. In a cohort comparing irritable/emotionally dysregulated ADHD children ($n = 105$) to non-irritable ADHD peers ($n = 395$), the former group showed a higher risk of oppositional defiant symptoms [44]. Therefore, both inhibitory control and emotion control are involved in the emergence and development of oppositional defiant symptoms.

Associations between socio-functional competences and ADHD symptoms

According to Table 2, some indicators from PEOPSS and Early Childhood Communication Interest Questionnaire are associated with symptoms of ADHD. Attention deficit was related to Pioneership and Expressiveness, hyperactivity/impulsivity was linked to Self-management, Pioneership and communication interests, while oppositional defiant symptoms were associated with Self-management and communication interests.

Previous research indicated significant differences in EF between Chinese and American preschoolers [45]. EF profiles may vary across cultures due to several interrelated factors. Cultural values shape parenting practices and educational goals, which in turn prioritize and reinforce different cognitive and behavioral skills. Furthermore, culturally specific experiences and skills trained from an early age can lead to differential strengths in neural circuits supporting distinct aspects of EF. Therefore, the assessment of EF and its manifestation in ADHD cannot be divorced from the cultural context in which a child develops. The PEOPSS tailored to the Chinese social and cultural context, provides a scientific and effective measurement tool for the comprehensive assessment of young children's social skills. The internal consistency reliability of the questionnaire and each factor ranged from 0.85 to 0.93, the retest reliability ranged from 0.70 to 0.77 over a four-week period, and the correlation with the validity scale ranged from 0.31 to 0.54. Pioneership operationalizes a preschoolers' capacity for prospective thinking–anticipating future scenarios and initiating preparatory actions. Expressiveness reflects the child's active communication of emotions and thoughts. Self-management refers to the child's ability to exert control over their own emotions and behaviors. During the preschool years, the focus of social skill development revolves around sharing and communicating, interacting with peers in a positive way, cooperating, and regulating emotions. Communication interests in early childhood are examined in terms of young children's primitive declarative pointing to things with their fingers, eye-to-eye gaze, and pretend play skills. These skills are associated with the differentiation of ADHD symptoms and future development.

Figure 1 demonstrates significant associations between attention deficits and Fine motor, Motor coordination, and the total score on the DCDQ-C. Previous studies showed that a considerable proportion of preschoolers with ADHD also have motor deficits that manifest as difficulties with fine and gross motor movements and motor control. These deficits negatively impact activities

of daily living—such as writing, using utensils, and tying shoelaces—often causing children to lag behind their peers [46, 47]. In particular, research found that children with ADHD exhibit poorer performance on all fine motor coordination tasks. Notably, lower performance was associated with symptoms of inattention rather than hyperactivity/impulsivity [48].

The context of the COVID-19 pandemic that was present during the period of data collection must be considered when interpreting these findings. The heightened levels of family stress, limited social interactions and potential alterations in healthcare-seeking and referral patterns, may have exacerbated children's behavioral symptoms and influenced parental reporting. Although the core neurocognitive relationships we investigated are considered fundamental and enduring, the generalizability of our specific findings to more stable periods merits future investigation.

Limitations and future directions

Several limitations are important to note. First, given the considerable challenges in validly assessing EF in preschoolers, our data were derived from caregiver reports of children's observable behaviors, rather than from child performance-based tasks. While individuals, including children, can report on their own experiences in unique ways, preschool children have limited language comprehension and self-insight, making their survey responses less reliable than those of adults. However, this single-informant approach may introduce potential measurement bias. Future studies should incorporate objective standardized neuropsychological tests, longitudinal designs, and multiple informants to obtain a more accurate and context-sensitive understanding of children's functioning [24]. Second, the sample's sex imbalance may limit generalizability to female populations. Our sample included a significantly higher number of male participants. This may be attributed to the greater prevalence in boys of hyperactive behaviors, which are often readily observable and potentially disruptive. However, since previous research indicates that EF deficits are comparable between girls and boys with ADHD [49, 50], the imbalance is unlikely to have substantially biased the observed relationships within our sample. Nonetheless, future research should recruit more balanced cohorts to ensure findings are generalizable across genders. Third, we did not systematically assess parental stress or mental health, which are potential confounders. Distressed parents may rate their children's behaviors more negatively, potentially inflating the observed associations. As a standardized measure, the BRIEF-P incorporates quality control procedures specifically designed to reduce such bias, including the calculation of an Inconsistency score and a Negativity score from the original ratings. Future research should incorporate standardized assessments of parental psychological status to better disentangle children's inherent behavioral characteristics from state-dependent biases in parent reports, thereby yielding more robust conclusions. Finally, our participant recruitment was confined to a single urban hospital in Zhejiang province. The findings may not be fully generalizable to children from rural areas, different socioeconomic backgrounds, or non-clinical settings, which should be considered when interpreting the results.

Conclusion

In summary, ADHD symptoms in preschoolers are associated with deficits in EF, motor coordination, social skills, and communication interests. Our findings underscore the strong discriminative validity of EF measures for identifying ADHD symptoms within the Chinese preschool population. Early detection of EF deficits contributes to a comprehensive assessment of ADHD symptoms and optimizes personalized intervention efficacy.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12887-025-06465-z.

Supplementary Material 1.

Acknowledgements

We would like to express our deepest gratitude to all children, parents that took part in this study.

Authors' contributions

Yi Gao and Guojuan Chen: Original draft, data curation; Jiangping Wang, and Cuiwei Yang: Data collection; Rongwang Yang: Data collection, review and editing. Kewen Jiang: Conceptualization, supervision and editing.

Funding

Kewen Jiang was supported by the National Natural Science Foundation of China (81871012 and 81571263). Guojuan Chen was supported by the Science and Technology Plan of Zhejiang Provincial Department of Health (2022519065). Cuiwei Yang was supported by the Natural Science Foundation of Zhejiang Province (LY20H090015).

Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

This study was approved by the Institutional Research Ethics Board of Children's Hospital, Zhejiang University School of Medicine (Approval No. 2023-IRB-0318-P-01), and complied with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. Given the retrospective nature of this study, the requirement for informed consent was waived. Clinical trial number: not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 11 May 2025 / Accepted: 15 December 2025
Published online: 08 January 2026

References

1. Wolraich ML, Hagan JF Jr, Allan C, et al. Clinical Practice Guideline for the Diagnosis, Evaluation, and Treatment of Attention-Deficit/Hyperactivity Disorder in Children and Adolescents. Pediatrics. 2019;144(4):e20192528. https://doi.org/10.1542/peds.2019-2528.
2. Thomas R, Sanders S, Doust J, Beller E, Glasziou P. Prevalence of attention-deficit/hyperactivity disorder: a systematic review and meta-analysis. Pediatrics. 2015;135(4):e994–1001.
3. Danielson ML, Bitsko RH, Ghandour RM, Holbrook JR, Kogan MD, Blumberg SJ. <Emphasis Type="Bold">Prevalence of parent-reported ADHD diagnosis and associated treatment among U.S. children and adolescents, 2016</Emphasis>. J Clin Child Adolesc Psychol. 2018;47(2):199–212.
4. Wang T, Liu K, Li Z, Xu Y, Liu Y, Shi W, et al. Prevalence of attention deficit/hyperactivity disorder among children and adolescents in China: a systematic review and meta-analysis. BMC Psychiatry. 2017;17(1):32.
5. Jensen CM, Steinhausen HC. Comorbid mental disorders in children and adolescents with attention-deficit/hyperactivity disorder in a large nationwide study. Atten Defic Hyperact Disord. 2015;7(1):27–38.
6. Ollendick TH, Jarrett MA, Grills-Taquechel AE, Hovey LD, Wolff JC. Comorbidity as a predictor and moderator of treatment outcome in youth with anxiety, affective, attention deficit/hyperactivity disorder, and oppositional/conduct disorders. Clin Psychol Rev. 2008;28(8):1447–71.
7. Spira EG, Fischel JE. The impact of preschool inattention, hyperactivity, and impulsivity on social and academic development: a review. J Child Psychol Psychiatry Allied Discip. 2005;46(7):755–73.
8. Childress AC, Stark JG. Diagnosis and treatment of Attention-Deficit/Hyperactivity disorder in preschool-aged children. J Child Adolesc Psychopharmacol. 2018;28(9):606–14.
9. Willcutt EG. The prevalence of DSM-IV attention-deficit/hyperactivity disorder: a meta-analytic review. Neurotherapeutics: J Am Soc Experimental Neurother. 2012;9(3):490–9.
10. Canals J, Morales-Hidalgo P, Jané MC, Domènech E. ADHD prevalence in Spanish preschoolers: comorbidity, socio-demographic factors, and functional consequences. J Atten Disord. 2018;22(2):143–53.
11. Rajaprakash M, Leppert ML. Attention-Deficit/Hyperactivity disorder. Pediatr Rev. 2022;43(3):135–47.
12. Friedman NP, Robbins TW. The role of prefrontal cortex in cognitive control and executive function. Neuropsychopharmacology: Official Publication Am Coll Neuropsychopharmacol. 2022;47(1):72–89.
13. Moriguchi Y, Hiraki K. Prefrontal cortex and executive function in young children: a review of NIRS studies. Front Hum Neurosci. 2013;7:867.
14. Funahashi S. Neuronal mechanisms of executive control by the prefrontal cortex. Neurosci Res. 2001;39(2):147–65.
15. Diamond A. Executive functions. Annu Rev Psychol. 2013;64:135–68.
16. Sadozai AK, Sun C, Demetriou EA, Lampit A, Munro M, Perry N, Boulton KA, Guastella AJ. Executive function in children with neurodevelopmental conditions: a systematic review and meta-analysis. Nat Hum Behav. 2024;8(12):2357–66.
17. Sjöwall D, Bohlin G, Rydell AM, Thorell LB. Neuropsychological deficits in preschool as predictors of ADHD symptoms and academic achievement in late adolescence. Child Neuropsychology: J Normal Abnorm Dev Child Adolescence. 2017;23(1):111–28.
18. Hawkey EJ, Tillman R, Luby JL, Barch DM. Preschool executive function predicts childhood Resting-State functional connectivity and Attention-Deficit/Hyperactivity disorder and depression. Biol Psychiatry Cogn Neurosci Neuroimaging. 2018;3(11):927–36.
19. Gioia GA, Isquith PK, Guy SC, Kenworthy L, Baron IS. Test review: behavior rating inventory of executive function. Child Neuropsychol. 2000;6(3):235–8.
20. Isquith PK, Gioia GA, Espy KA. Executive function in preschool children: examination through everyday behavior. Dev Neuropsychol. 2004;26(1):403–22.
21. Duku E, Vaillancourt T. Validation of the BRIEF-P in a sample of Canadian preschool children. Child Neuropsychology: J Normal Abnorm Dev Child Adolescence. 2014;20(3):358–71.
22. Skogan AH, Zeiner P, Egeland J, Urnes AG, Reichborn-Kjennerud T, Aase H. Parent ratings of executive function in young preschool children with symptoms of attention-deficit/-hyperactivity disorder. Behav Brain Functions: BBF. 2015;11:16.
23. Landis TD, Garcia AM, Hart KC, Graziano PA. Differentiating symptoms of ADHD in preschoolers: the role of emotion regulation and executive function. J Atten Disord. 2021;25(9):1260–71.
24. Ceruti C, Marzocchi GM. Beyond the Overlap: Understanding the Empirical Association Between ADHD Symptoms and Executive Function Impairments in Questionnaire-Based Assessments. Children (Basel). 2025;12(8):970. https://doi.org/10.3390/children12080970.
25. Willcutt EG, Doyle AE, Nigg JT, Faraone SV, Pennington BF. Validity of the executive function theory of attention-deficit/hyperactivity disorder: a meta-analytic review. Biol Psychiatry. 2005;57(11):1336–46.
26. Willoughby MT, Wylie AC, Blair CB. Using repeated-measures data to make stronger tests of the association between executive function skills and attention deficit/hyperactivity disorder symptomatology in early childhood. J Abnorm Child Psychol. 2019;47(11):1759–70.
27. Peisch V, Ferrara E, Gourdet G, Zimon M, Mastrangelo C, Arnett A. Temperament and executive functioning correlates of ADHD symptom severity during early versus middle childhood. J Child Psychol Psychiatry Allied Discip. 2025;66(5):752–62.
28. Gioia GA, Espy KA, Isquith PK. BRIEF-P: behavior rating inventory of executive function–preschool version. Psychological Assessment Resources (PAR); 2003.
29. Zhang HF, Shuai L, Zhang JS, Wang YF, Lu TF, Tan X, et al. Neuropsychological profile related with executive function of Chinese preschoolers with Attention-Deficit/Hyperactivity disorder: neuropsychological measures and behavior rating scale of executive function-preschool version. Chin Med J. 2018;131(6):648–56.
30. Tseng MH, Fu CP, Wilson BN, Hu FC. Psychometric properties of a Chinese version of the developmental coordination disorder questionnaire in community-based children. Res Dev Disabil. 2010;31(1):33–45.
31. Swanson JM, Kraemer HC, Hinshaw SP, Arnold LE, Conners CK, Abikoff HB, Clevenger W, Davies M, Elliott GR, Greenhill LL, et al. Clinical relevance of the primary findings of the MTA: success rates based on severity of ADHD and ODD symptoms at the end of treatment. J Am Acad Child Adolesc Psychiatry. 2001;40(2):168–79.
32. Kofler MJ, Irwin LN, Soto EF, Groves NB, Harmon SL, Sarver DE. Executive functioning heterogeneity in pediatric ADHD. J Abnorm Child Psychol. 2019;47(2):273–86.
33. van Lieshout M, Luman M, Twisk JW, Faraone SV, Heslenfeld DJ, Hartman CA, Hoekstra PJ, Franke B, Buitelaar JK, Rommelse NN, et al. Neurocognitive predictors of ADHD outcome: a 6-Year Follow-up study. J Abnorm Child Psychol. 2017;45(2):261–72.
34. Irwin LN, Soto EF, Chan ESM, Miller CE, Carrington-Forde S, Groves NB, Kofler MJ. Activities of daily living and working memory in pediatric attention-deficit/hyperactivity disorder (ADHD). Child Neuropsychology: J Normal Abnorm Dev Child Adolescence. 2021;27(4):468–90.
35. McAuley T, Crosbie J, Charach A, Schachar R. The persistence of cognitive deficits in remitted and unremitted ADHD: a case for the state-independence of response inhibition. J Child Psychol Psychiatry Allied Discip. 2014;55(3):292–300.
36. Brocki KC, Eninger L, Thorell LB, Bohlin G. Interrelations between executive function and symptoms of hyperactivity/impulsivity and inattention in preschoolers: a two year longitudinal study. J Abnorm Child Psychol. 2010;38(2):163–71.
37. Groves NB, Chan ESM, Marsh CL, Gaye F, Jaisle EM, Kofler MJ. Does central executive training and/or inhibitory control training improve emotion regulation for children with attention-deficit/hyperactivity disorder? A randomized controlled trial. Front Psychiatry. 2022;13:1034722.
38. Graziano PA, Garcia A. Attention-deficit hyperactivity disorder and children's emotion dysregulation: a meta-analysis. Clin Psychol Rev. 2016;46:106–23.
39. Bunford N, Evans SW, Langberg JM. Emotion dysregulation is associated with social impairment among young adolescents with ADHD. J Atten Disord. 2018;22(1):66–82.
40. Sjöwall D, Roth L, Lindqvist S, Thorell LB. Multiple deficits in ADHD: executive dysfunction, delay aversion, reaction time variability, and emotional deficits. J Child Psychol Psychiatry Allied Discip. 2013;54(6):619–27.
41. Sjöwall D, Backman A, Thorell LB. Neuropsychological heterogeneity in preschool ADHD: investigating the interplay between cognitive, affective and motivation-based forms of regulation. J Abnorm Child Psychol. 2015;43(4):669–80.
42. Schoemaker K, Mulder H, Deković M, Matthys W. Executive functions in preschool children with externalizing behavior problems: a meta-analysis. J Abnorm Child Psychol. 2013;41(3):457–71.

43. Bohlin G, Eninger L, Brocki KC, Thorell LB. Disorganized attachment and inhibitory capacity: predicting externalizing problem behaviors. J Abnorm Child Psychol. 2012;40(3):449–58.
44. Ambrosini PJ, Bennett DS, Elia J. Attention deficit hyperactivity disorder characteristics: II. Clinical correlates of irritable mood. J Affect Disord. 2013;145(1):70–6.
45. Lan X, Legare CH, Ponitz CC, Li S, Morrison FJ. Investigating the links between the subcomponents of executive function and academic achievement: a cross-cultural analysis of Chinese and American preschoolers. J Exp Child Psychol. 2011;108(3):677–92.
46. Athanasiadou A, Buitelaar JK, Brovedani P, Chorna O, Fulceri F, Guzzetta A, Scattoni ML. Early motor signs of attention-deficit hyperactivity disorder: a systematic review. Eur Child Adolesc Psychiatry. 2020;29(7):903–16.
47. Fliers E, Rommelse N, Vermeulen SH, Altink M, Buschgens CJ, Faraone SV, Sergeant JA, Franke B, Buitelaar JK. Motor coordination problems in children and adolescents with ADHD rated by parents and teachers: effects of age and gender. J Neural Transmission (Vienna Austria: 1996). 2008;115(2):211–20.
48. Fenollar-Cortés J, Gallego-Martínez A, Fuentes LJ. The role of inattention and hyperactivity/impulsivity in the fine motor coordination in children with ADHD. Res Dev Disabil. 2017;69:77–84.
49. Seidman LJ, Biederman J, Monuteaux MC, Valera E, Doyle AE, Faraone SV. Impact of gender and age on executive functioning: do girls and boys with and without attention deficit hyperactivity disorder differ neuropsychologically in preteen and teenage years? Dev Neuropsychol. 2005;27(1):79–105.
50. Skogli EW, Teicher MH, Andersen PN, Hovik KT, Øie M. ADHD in girls and boys–gender differences in co-existing symptoms and executive function measures. BMC Psychiatry. 2013;13:298.

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